Exploring rare disease realities: a systematic literature review on harnessing social media patient-generated data

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Introduction

Background

Rare diseases affect roughly 400 million people world-wide, and 30 million in Europe [1]. Intangible costs and personal aspects for patients and their families are rarely accounted for, as most studies focus on easy-to-measure metrics. On the other hand, social media play an important role for these people who can easily feel isolated and seek both support and advice online [2]. With 63% of the world's population using the internet in 2021 [3], and 93.4% of internet users being on social media [14], social media are growing into really rich sources of information not only for individuals – patients and families – but also for public health policymakers.

Objective

The goal of this study is to systematically review the peer-reviewed academic literature on the use of social media data generated by rare disease patients and families. More precisely, our study will be focused on genetic and congenital rare diseases. In this review, we examine how social media has been used to generate new knowledge, and the types of research questions that have been answered through social data mining. Moreover, we explore what types of methods, data sources, and data types are used, with a focus on major results, uncovered needs and opportunities in rare-disease-oriented AI research using social media.

Methods

Search strategy

Expanding on Miller et al.'s [4] search strategy, our study, focusing on user-generated data, includes 'content analysis' publications from their review. Utilizing their search strings, we queried five databases (PubMed, Embase, CINAHL, PsycINFO, and Web of Science) from November 2020 to May 2023.

Inclusion criteria

- Focused on rare non-communicable diseases
- Used social media data as a primary source of data
- Published in English in a peer-reviewed journal between January 2004 and May 2023

Variables used for manual annotation

- Metadata associated with the publication: title, DOI, authors, journal, year of publication
- Objectives of the publication: study goal as stated and disease(s) studied
- Data source: social media from which data is extracted, language of the posts extracted, number of posts included, number of users included, temporal range (i.e. on what period of time were the messages studied posted)
- Methods and results as stated
- AI model: when relevant purpose of the model, type of model used, performance scores used and their values

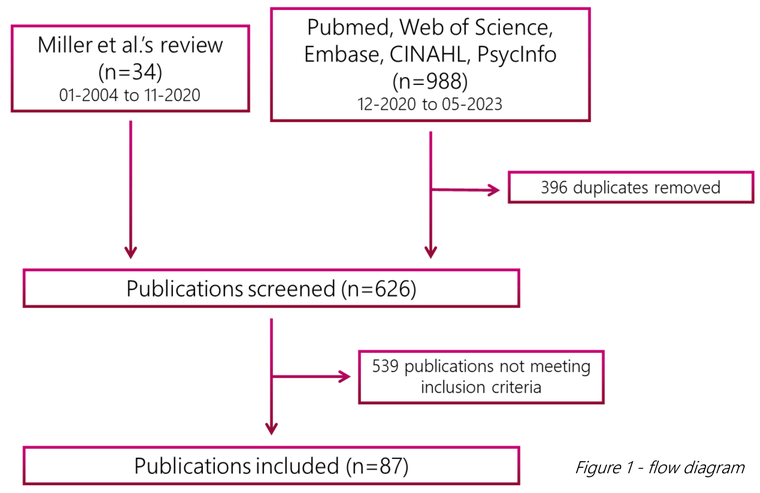


Figure 1 - flow diagram

Results

Number of publications meeting our inclusion criteria

- Before 2014: less than 1 per year
- Between 2014 and 2020: less than 5 per year
- 2021: 21
- 2022: 29

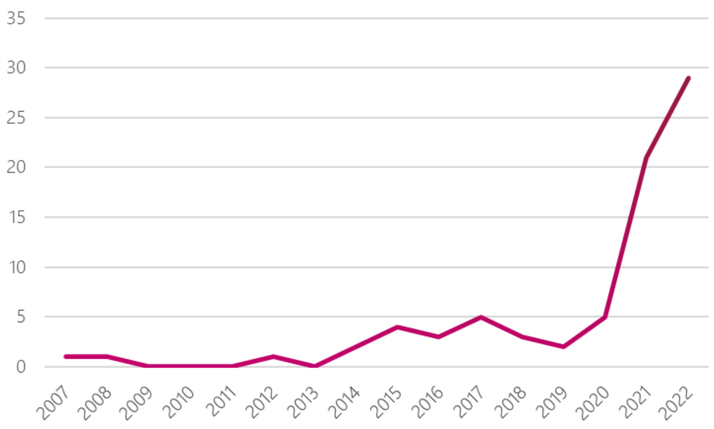


Figure 2 - number of included publications by year of publication

Type of content studied

In the vast majority of the studies (n=81; 93.1%), the contents analysed were in English. Other languages are French (n=3; 3.4%), Spanish (n=2; 2.3%), and Danish, Dutch, German and Chinese (only once each).

The number of studies using video has increased in the last three years: among the eighteen publications focused on video, all but one were published after 2020.

Similarly, the four publications focused on images were all published after 2021.

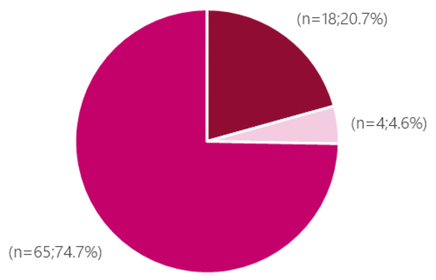


Figure 3 - type of content analysed

Categories are not exclusive	Number of studies	Percentage of studies
	27	31.0
	27	31.0
	23	26.4
	18	20.7
	17	19.5
	11	12.6
	9	10.3
ooo	4	4.6

Table 1 – number and proportion of publications by social media

Study goals

	Number of studies	Percentage of studies
Patients and their families	39	44.8
Communities and groups	18	20.7
Scientific relevance/quality of online content	18	20.7
Comparison between patients' and professionals' contents	3	3.4
Other	9	10.3

Table 2 - number and proportion of publications by study goal

Rare diseases studied

	Number of studies	Percentage of studies
Single rare disease	60	69.0
Group of rare diseases	22	25.3
Comparison of rare diseases	1	1.1
Rare diseases in general	4	4.6

Table 3 - number and proportion of publications by focus

Besides Sickle Cell Disease (n=6; 6.9%), Cleft Lip/Palate (n=5; 5.7%), Cystic Fibrosis (n=4; 4.6%) - which represent almost a fifth of the publications (17.2%), the other diseases were mentioned in only one (n=19; 21.8%) or two (n=5; 5.7%) publications each.

Patients' and caregivers' insights

- Aimed at better understanding patients' or caregivers' global experience.
- Did not focus on only one aspect and could include insights about the experience, the symptoms, the quality of life, the care and treatment pathways, the diagnosis, the seeking and giving of advice, the disease awareness.

Insights about specific experiences

Highlighted the possible differences between the aspects of disease reported in the literature and those discussed in patient conversations.

Examples:

- *Understand feeding difficulties of children following esophageal atresia/tracheo-esophageal fistula repair and the impact of these feeding difficulties on parents* [5]
- *Evaluate bullying in individuals with Cleft lip/palate* [6]

Insights about treatments

Focused on patients and caregivers' perspectives on treatments, or new treatments.

Examples:

- *Study the impact on parents' hopes and expectations of a new treatment for Duchenne muscular dystrophy or spinal muscular atrophy type 2* [7]
- *Understand barriers to Hydroxyurea for Sickle Cell Disease patients* [8]
- *Perspectives on heavy surgical treatments such as tooth removal for Lesch-Nyhan disease patients* [9]

Insights about genetic risk and testing

Aimed at understanding how people with a known genetic risk reacted to it and evaluated the need to be tested and the stress induced.

Examples:

- *Find out how people with a family history of Amyotrophic Lateral Sclerosis make sense of and negotiate genetic risk* [10]
- *Analyse health forums posts about genetic testing of Huntington's disease* [11]

Insights about the impact of Covid-19

- Studied the consequences of Covid-19 on patients' [12–14] or parents' [15] experiences and concerns.
- Focused on the disruption caused by the global crisis for people suffering from rare rheumatologic, cardiac, pulmonary, or esophageal diseases.

Insights about support on social media

Assessed the support given and received through social media.

Example:

- *Study of messages from a Huntington's disease online support group using a social support framework: informational and emotional support were the most offered, compared to esteem support and tangible assistance* [16]

Methods used to conduct the studies

Surprisingly, automated methods were used in only seven studies, all published after 2020: two studies in 2020/2021 and five in 2022/2023.

Topic identificaiton (n=4)	• Regex [17] • Comparison LDA and Top2Vec with six different embeddings [18] • Comparison LDA and TF-IDF [19] • BERTopic [20]	• Coherence (0.642) [17] • F1 score (0.902) [19]
Sentiment analysis (n=2)	Comparison of LSTM with Naive Bayes Classifier, SVC, CNN [19]	F1 score (0.916) [19]
Other tasks (n=2)	Text generation: GPT-2 [21]	Clarity (3.58 AI vs 3.34 human) & quality (3.57 AI vs 3.3 human) [21]

Table 4 - AI-assisted tasks, models and performance scores used

	Total	Studies with AI methods	Studies without AI methods
Number of studies	57	7	50
Mean	5,310	**33,201**	1,405
Minimum	16	317	16
Maximum	120,738	**120,738**	13,089

Table 5 - number of posts included in the studies

Among the publications which indicate it (n=42; 48.3%), almost 70% had a time range of less than three years (n=29; 33.3%). Among the publications using AI methods, three had a temporal range of five years or more, accounting for half of the publications with a temporal range of five years or more.

Conclusions

Some limitations may be inherent to the selection and the analysis of the publications, although we used the methodology previously published by Miller et al. [4] and designed the protocol to be as objective as possible.

Social media mining in the rare disease area is still a research domain that has not benefited from recent advances in AI.

The use of social media user-generated data could provide patients, health professionals, and researchers with information that is not accessible otherwise. It could also help tackle the issues arising from the low prevalence of the diseases, the need for support and the possible isolation of the patients. In other words, real-world data from social media should play an important role in rare disease research. Our results suggest that it could be increasingly developed and should employ innovative methods and approaches.

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